



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

Thesis Title.

MASTER THESIS

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1 Introduction

Teleoperation for robotic dexterous manipulation is a very difficult task. Some reasons include Operator Hand shape differences Manipulator hand shape differences We see that there are approaches to perform manipulation using RL, as the policies can learn how to maximize the contact forces, etc. with the object. However, many of these approaches use object-specific reward functions or train a single policy for a single demonstration. Using a policy for teleoperation for single task would therefore be infeasible as the policy is fitted to one instance.

2 Methods

We use a method of multi-demo, multi-stage curriculum learning to get good multi-demo performance, and validate it using a task of demonstrations for a single task.

3 Results

ManipTrans Baseline

Policy trained on a single demo, running other demonstration performance

Policy trained on multiple demos running other demonstrations

4 Discussion

5 Conclusion

6 References

7 Appendix